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The impact of virtual care on health-related quality of life in children with diabetes mellitus: A systematic review protocol

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Manuscripts

The impact of virtual care on health-related quality of life in children with diabetes mellitus: A systematic review protocol

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ABSTRACT

Introduction: Diabetes mellitus is the most common endocrine disorder in children, and the prevalence of pediatric Type 1 and Type 2 diabetes continues to rise around the world. Diabetes care pivoted to virtual care with the COVID-19 pandemic-driven distancing measures. Yet, the impact of virtual care on the quality of life of children living with diabetes remains unclear. This systematic review aims to measure the health-related quality of life and metabolic health impacts of virtual diabetes care.

Methods and analysis: We will search MEDLINE, Embase, EMCare, PsycInfo, Web of Science, and the grey literature for eligible studies. We will screen title, abstract, and full-text papers for potential inclusion and assess the risk of bias and the overall confidence in the evidence using GRADE. A meta-analysis will be conducted if two studies report similar populations, study designs, methods, and outcomes.

This systematic review will summarize the health-related quality of life outcomes for virtual diabetes care delivery models. Findings from this review will identify successful interventions that may be potentially scalable for broader use. We will disseminate the results of this systematic review via publication in a peer-reviewed journal.

Ethics and dissemination: No ethics approval is required for this systematic review protocol as it does not include patient data. The systematic review will be published in a peer-reviewed journal and presented at international conferences.

PROSPERO Registration Number: CRD42021235646

Keywords: Virtual, eHealth, mHealth, pediatric, diabetes mellitus, education, care

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ARTICLE SUMMARY

Strengths and limitations of this study:

- This systematic review protocol includes a detailed reporting of the methods to assess health-related quality of life outcomes of the virtual delivery of diabetes care and education in pediatric diabetes patients, to guide future directives in implementing virtual care.
- If there is high heterogeneity between studies, it may not be possible to perform a meta-analysis.
- The sample includes both types of diabetes (T1DM and T2DM), resulting in less clinically relevant conclusions if the sample size per study is small.

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INTRODUCTION

Diabetes mellitus is the most common endocrinopathy in children and currently, over 1.1 million children and youth are living with type 1 diabetes mellitus (T1DM) (1, 2). While the global prevalence of type 2 diabetes mellitus (T2DM) has not been fully measured, the prevalence rates are estimated to be up to 5,330/100,000 children in 2013 (3). The prevalence of type 2 diabetes mellitus in children is steadily rising in parallel with the rising obesity rates globally (1, 4).

Over the past few decades, the outlook for diabetes outcomes was transformed due to advances in understanding the importance of tight glycemic control and aggressive screening for comorbidities (5). Current goals of diabetes care focus on achieving and maintaining glycemic control to reduce the risk of diabetes-related complications (6-10).

In addition to the development of new insulins and insulin delivery systems, diabetes care has been transformed through the development of outpatient-based multidisciplinary care teams (7, 10). Multiple health care professionals, including a mix of nurses, dietitians, social workers, child life specialists, psychologists, and physicians, are typically involved in the care for patients and families, and this approach has achieved essential improvements in diabetes outcomes (6, 7, 11, 12). The North American guidelines and others recommend that these clinics occur quarterly (6, 7, 13).

While this approach to care is quite beneficial, it is labor-intensive, costly, and can be disruptive to work/school time, with a negative impact on health-related quality of life (HRQOL) due to absences from work and school (11, 14, 15).

Since the start of the COVID-19 pandemic, multidisciplinary-based diabetes care needed to shift swiftly to virtual platforms due to social distancing measures to protect patients and manage finite healthcare resources (16, 17). While it is unclear whether virtual care has led to a negative

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impact on diabetes care and outcomes in the short term, the full implications of this shift in care delivery are only emerging (17).

Virtual care for diabetes during the COVID-19 pandemic is building on existing trends of remote delivery of care. Web-based (eHealth) and mobile-based (mHealth) platforms link patients and families with healthcare providers (18). These modalities have been surging over the past few years in healthcare. They are potentially quite promising as they are relatively inexpensive, scalable, with successful reported patient engagement, satisfaction with communication with healthcare teams, and the facilitation of access to specialized care and diabetes monitoring (19, 20).

However, current evidence has focused on how these interventions maintain glycemic control and metabolic health outcomes, yet this improvement in diabetes control often comes as a trade-off with HRQOL (8). The eHealth and mHealth approaches increase patient responsibilities with a need for familiarity with these care modules' technical aspects (20, 21). There is currently no existing evidence synthesis of the impact of virtual interventions on HRQOL in children living with diabetes.

This systematic review aims to determine the effect of virtual models of diabetes care and education on HRQOL in children living with diabetes. We will also assess glycemic control, adverse events, user satisfaction, and perceptions of virtual care acceptability and feasibility.

Objectives

This systematic review aims to determine the effectiveness of existing eHealth and mHealth interventions in maintaining or improving HRQOL, glycemic control, overweight and obesity, and adverse events in children living with diabetes. Also, we aim to determine user satisfaction, and their perceptions of the acceptability and feasibility of virtual care modalities.

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METHODS

The methodology for this protocol has been reported based on the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) guideline (22, 23) (Supplementary File 1). This protocol has been registered with PROSPERO (Registration Number: CRD42021235646).

Search Strategy

This review's search strategies will be developed by a Senior Health Sciences Librarian (LB), who has expertise in designing systematic review search strategies. A sample Ovid MEDLINE search strategy is reported in Table 1. Our searches will run on the following platforms: Ovid MEDLINE, Ovid Embase (Elsevier), Ovid EMCare (Elsevier), PsycInfo, and Web of Science (Clarivate). The grey literature searches will be done using ProQuest Dissertations and Theses A&I, and ClinicalTrials.gov.

Table 1: Ovid MEDLINE Search Strategy

#	Searches
1	Child/
2	child*.mp.
3	Adolescent/
4	adolescen*.mp.
5	Pediatrics/
6	pediatric*.mp.
7	paediatric*.mp.
8	teen*.mp.
9	preadolescen*.mp.
10	preteen*.mp.
11	or/1-10
12	Diabetes Mellitus, Type 2/ or Diabetes Mellitus, Type 1/ or Diabetes Mellitus/
13	(T1D or T1DM or T2DM or T2D or NIDDM or IDDM).mp.

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14	diabet*.mp.
15	or/12-14
16	Telemedicine/
17	telemedicine.mp.
18	tele-medicine.mp.
19	mhealth.mp.
20	ehealth.mp.
21	virtual health.mp.
22	Mobile Applications/
23	teleconsult*.mp.
24	web-based.mp.
25	mobile-based.mp.
26	Social media/
27	social media.mp.
28	twitter.mp.
29	telemonitor*.mp.
30	Remote Consultation/
31	iphone.mp.
32	computers/ or computers, handheld/ or smartphone/
33	smart phone*.mp.
34	smartphone*.mp.
35	computer*.mp.
36	laptop*.mp.
37	((remote or web or virtual or online) adj2 (consult* or appointment*)).mp.
38	((web or virtual or online) adj2 (education or communit* or platform*)).mp.
39	ipad.mp.
40	(android adj3 (tablet* or device* or phone*)).mp.
41	or/16-40
42	Comparative Study/
43	comparative stud*.mp.
44	cohort studies/ or prospective studies/ or retrospective studies/

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45	cohort stud*.mp.
46	Observational Study/
47	observation* stud*.mp.
48	nonrandom*.mp.
49	non-random*.mp.
50	Randomized Controlled Trial/ or Randomized Controlled Trials as Topic/
51	randomized controlled trial*.mp.
52	Random Allocation/
53	random*.mp.
54	Clinical Trial/
55	clinical trial.mp.
56	retrospective.mp.
57	prospective.mp.
58	or/42-57
59	11 and 15 and 41 and 58
60	remove duplicates from 59

Inclusion Criteria

We will include studies that recruited boys and girls ages 2-18 years, from all ethnicities and diagnosed with T1DM or T2DM. Children at all pubertal stages based on the Tanner staging (24, 25), glycemic control levels (using Glycated Hemoglobin A1c, HbA1c), and Body Mass Index (BMI) z-scores will be included. Children treated with insulin injections of all insulin types and regimens (multiple daily injection regimens, twice-daily regimens, insulin mixes) will be included. We will also have children on insulin pump therapy regardless of pump type and those using glucometers or continuous glucose sensors as applicable.

We will include studies of all geographic locations, languages, and publication timing.

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The eligible study designs to be included in this systematic review are randomized and non-randomized comparative studies. We will exclude protocol papers, case reports, and commentaries. Eligible studies which are not published in English will be translated using Google Translate.

Children with co-morbid autoimmune conditions (for example, Hashimoto's thyroiditis, Graves' disease, celiac disease), as well as children with diabetes-related complications including neuropathy, nephropathy, and retinopathy will be included.

This review will include interventions that deliver diabetes care via modalities described as eHealth and mHealth. Diabetes education and care refer to personalized patient engagement to monitor and promote glycemic control and limit comorbidities (7, 10). eHealth refers to the Internet, virtual reality, or digital gaming applications that allow users to monitor, manage or learn more about their health through video, text, or interactive learning media (26-28). mHealth refers to portable or wireless applications such as text messaging, mobile-compatible applications (apps), wearable devices, or the use of social media (26, 27, 29). These applications also allow for user engagement with health monitoring, management, and learning (28-30).

Comparator groups will be reported if available. These groups will be delivering treatment as usual; this approach includes diabetes education and care delivery via in-person clinic visits with diabetes healthcare teams. These visits include a review of glucose trends in logbooks or sensor downloads, a review of insulin regimens for injectors and pump settings for children on insulin pumps, dietary guidance, physical activity recommendations, and emotional and mental health screening and supports (14, 18). Also, information and education about lifestyle factors are included in these visits (14, 18).

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We will include studies that report on the HRQOL, measured by diabetes HRQOL questionnaires. These can consist of the Pediatric Quality of Life Inventory (PedsQL) (31, 32) or the Diabetes Quality of Life for Youth Questionnaire (DQOL-Y) (33). Other questionnaires, such as the proxy formats of general HRQOL questionnaires for parents and families and others, will be included.

Exclusion Criteria

We will exclude studies on boys and girls with other types of diabetes mellitus, including Maturity Onset Diabetes of the Young (MODY) and gestational diabetes mellitus, and cystic fibrosis-related diabetes, among others. Pregnant participants will also be excluded. Children receiving medications that can drive hyperglycemia e.g., steroids and immunosuppressants, will also be excluded.

OUTCOMES MEASURES

Critical Outcome

This systematic review's critical outcome includes the maintained or improved HRQOL, as measured by diabetes HRQOL questionnaires.

Important Outcomes

Important outcomes of this study include:

1) Maintained or improved glycemic control, as measured by HbA1c, logbook, or glucose sensor-based data, e.g., DexCom G5™ (34), FreeStyle Libre sensors (35) or others. While the target HbA1c level for people with diabetes is usually around 7-7.5%, goals that are adjusted to participants' needs will be accepted (6, 7, 36, 37).

2) Changes in body mass as measured by assessing BMI z-scores or BMI percentiles pre- and post-intervention changes.

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3) The number and severity of morbidities. These morbidities include hypoglycemia (glucose levels ≤ 3.9 mmol/L or 70 mg/dL) and hyperglycemia (glucose levels ≥ 13.3 mmol/L or 240 mg/dL). Also, the occurrence of ketosis, assessed via measurement of blood beta-hydroxybutyrate or urine acetoacetate and using manufacturer-specific reporting measures including (+) sign for urine ketones and units in mmol/L for blood ketones as applicable. We will also assess rates of diabetic ketoacidosis (defined as hyperglycemia with blood glucose >11 mmol/L or 200 mg/dL, venous blood gas-based pH <7.3 or serum bicarbonate <15 mmol/L, and the presence of ketones of ≥3 mmol/L in blood, moderate-large urine ketones) (6, 38). If no specific values reported the diagnosis of hypoglycemia, hyperglycemia, ketosis, and diabetic ketoacidosis, the above events would be included as a dichotomous outcome.

4) Patient and family satisfaction, and their perceptions of the acceptability and feasibility of the virtual intervention from relevant questionnaires.

Data Management

We will export search results onto Covidence (39) to remove duplicates and to complete data screening. Data abstraction will be conducted on Microsoft Excel. We will develop and pilot test the data abstraction form to ensure validity.

Study Selection

Two reviewers will independently assess the eligibility of titles, abstracts, and full-text articles identified in the database searches for potential eligibility and inclusion in this review. Disagreements at each stage will be resolved through discussion among the reviewers. A third reviewer will resolve persistent disagreements. The 2020 PRISMA flow diagram reported in Figure 1 will be utilized to track the process and be noted in the final paper (40, 41).

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Data Abstraction

Data from included full-text articles will be extracted independently by the two reviewers into a standardized data abstraction form. The reviewers will resolve any conflicts through discussion, and if necessary, a third reviewer will resolve any persisting discrepancies. We will abstract data on the population, interventions, controls, and results from relevant outcomes to determine whether studies can be pooled together for meta-analyses.

To obtain missing data for eligible papers, we will contact the Principal Investigators via e-mail.

Risk of Bias Assessment

Two reviewers will independently perform the risk of bias assessments of all randomized controlled trials using the Cochrane RoB 2.0 tool (42). Comparative, non-randomized studies will be assessed for risk of bias using the ROBINS-I tool (43). Any conflicts will be discussed, and a third reviewer will resolve any remaining disagreements.

GRADE assessment

To assess our overall confidence in the cumulative evidence quality, we will use the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach (44). Quality will be determined by study design, risk of bias, inconsistency, indirectness, imprecision, and publication bias (44). The quality of the cumulative evidence and magnitude of the effect will be utilized to determine the overall strength of the meta-analysis findings if it is feasible to do so.

Statistical Analysis

We will include a summary table of all eligible studies in our final review. A meta-analysis will be conducted if two or more studies of similar study design, populations, interventions, and outcomes are identified. We will report dichotomous outcomes as risk ratios or odds ratios and continuous variables as mean differences or standardized mean differences, with a 95%

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confidence interval. A detailed qualitative synthesis of the findings will be provided in a narrative table if a meta-analysis is not possible.

We will perform a random-effects model which accounts for within-study and between-study variability, as we expect high heterogeneity across studies. We will use the Review Manager (Revman 5.4) software to generate forest plots to represent the data graphically if appropriate (45).

Heterogeneity will be assessed using the I^2 statistic, and a cut-off value of >75% will be considerable heterogeneity (46). A sensitivity analysis will be performed by removing studies with a high risk of bias to assess the impact on meta-analysis results (47).

If appropriate, we will conduct subgroup analyses on the type of diabetes mellitus, as participants of T1DM and T2DM have fundamentally different management goals. We will also perform subgroup analyses on the age and sex of the participants and the mode of insulin administration (injections versus pumps).

If ten or more studies are included, we will generate a funnel plot and utilize Egger's test to assess small study effect/publication bias. Alternatively, we will report publication bias by considering the number of relevant conference abstracts that did not have published articles.

DISCUSSION

As virtual diabetes care and education have become viable alternatives in the COVID-19 era, understanding its effects on HRQOL outcomes in children with diabetes will guide future directives on implementing the digital delivery of diabetes care.

ETHICS AND DISSEMINATION

Ethics approval is not required as we will not recruit nor include any individual patient data in this systematic review. There are no other ethical and safety considerations. The results of our

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systematic review will be published in a peer-reviewed journal and presented at international conferences.

Author's Contributions: MCS is the guarantor. RR, MK, LB, UA, LT, and MCS were responsible for developing the research question, study conception, and design. MCS, LB, RR, MK, UA, and LT developed the search strategy and eligibility criteria. RR and MCS were responsible for drafting the manuscript. All authors reviewed, edited, and approved the final version of the manuscript.

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Data availability statement: There are no data in this work.

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Figure 1 Caption: Flow Diagram of Article Screening Process

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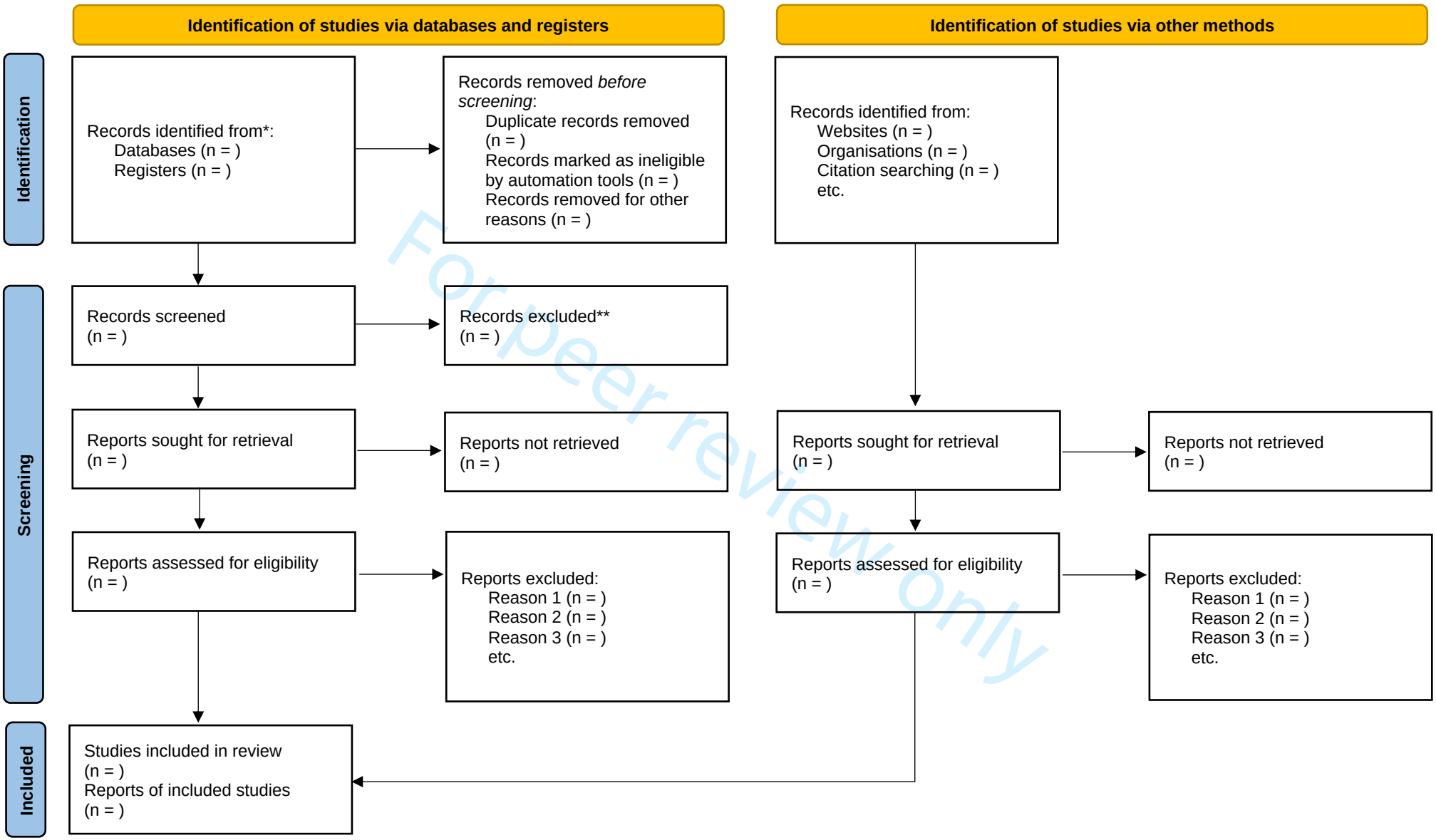
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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1-2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	57, 133-134
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	4-37
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	270-274
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	N/A
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	275-276
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	275-276
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	275-276
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	116-124

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	122-129
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	144-185
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	135-141
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	142
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	210-213
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	214-219
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	221-227
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	221-227
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	186-209
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	228-232

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	240-242
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	246-259
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	253-256
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	244-245
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☒	☐	257-259
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	233-238

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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BMJ Open

The impact of virtual care on health-related quality of life in children with diabetes mellitus: A systematic review protocol

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SCHOLARONE™
Manuscripts

The impact of virtual care on health-related quality of life in children with diabetes mellitus: A systematic review protocol

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ABSTRACT

Introduction: Diabetes mellitus is the most common endocrine disorder in children, and the prevalence of pediatric both Type 1 and Type 2 diabetes continue to rise globally. Diabetes care pivoted to virtual care with the COVID-19 pandemic-driven social distancing measures. Yet, the impact of virtual care on health-related quality of life of children living with diabetes remains unclear. This systematic review aims to measure the health-related quality of life and metabolic health impacts of virtual diabetes care.

Methods and analysis: We will search MEDLINE, Embase, EMCare, PsycInfo, Web of Science, and the grey literature for eligible studies. We will screen title, abstract, and full-text papers for potential inclusion and assess the risk of bias and the overall confidence in the evidence using GRADE. A meta-analysis will be conducted if two studies report similar populations, study designs, methods, and outcomes.

This systematic review will summarize the health-related quality of life outcomes for virtual diabetes care delivery models. We will disseminate the results of this systematic review via publication in a peer-reviewed journal.

Ethics and dissemination: No ethics approval is required for this systematic review protocol as it does not include patient data. The systematic review will be published in a peer-reviewed journal and presented at international conferences.

PROSPERO Registration Number: CRD42021235646

Keywords: Virtual, eHealth, mHealth, pediatric, diabetes mellitus, education, care

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ARTICLE SUMMARY

Strengths and limitations of this study:

- This systematic review protocol includes a detailed reporting of the methods to assess health-related quality of life outcomes of the virtual delivery of diabetes care and education in pediatric patients, and guide future policy directions in implementing virtual care in this population.
- The use of virtual care models and their impact on diabetes outcomes is a relatively novel area of care, and having data on their impact is critical to justify scalability
- If there is high heterogeneity, it will not be feasible to conduct a meta-analysis to estimate effect size

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INTRODUCTION

Diabetes mellitus is the most common endocrinopathy in children (1). Over 1.1 million children and youth are living with type 1 diabetes mellitus (T1DM) globally (2, 3). While the global prevalence of type 2 diabetes mellitus (T2DM) has not been fully measured, the prevalence rates are estimated to be up to 5,330/100,000 children in 2013 (4), and pediatric T2DM rates are steadily rising in parallel with its main driver, obesity (2, 5).

Diabetes care has been transformed over the past few decades due to the ability to achieve adequate glycemic control with the development of new insulins, insulin delivery systems, and glucose monitoring devices (6-12). In addition, outpatient-based multidisciplinary care have become the norm, and guidelines recommend that these clinics occur quarterly (6, 7, 10, 11, 13, 14). This care approach, while beneficial, is labor-intensive, costly, and can be disruptive to work/school time for children and families, with a negative impact on health-related quality of life (HRQOL) (13, 15, 16).

Since the start of the COVID-19 pandemic, a seismic shift in diabetes care took place; multidisciplinary care needed to shift swiftly to virtual platforms due to social distancing measures to protect patients and manage finite healthcare resources (17, 18). While it is unclear whether virtual care has had a negative impact on outcomes, the full implications of this shift in care delivery are starting to emerge (18).

Virtual care for diabetes during the COVID-19 pandemic is building on existing trends of increased access to remote care using web-based (eHealth) and mobile-based (mHealth) platforms. These platforms included telephone, Skype for Business, and Zoom that allow audio and video communications between patients, families, and healthcare providers (17, 19). The online platforms of continuous glucose monitoring devices, such as Dexcom CLARITY® and

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FreeStyle® Libre were leveraged for sharing accurate glycemic control and management data with healthcare providers (17, 18, 20, 21).

eHealth refers to internet, virtual reality, or digital gaming applications that allow users to monitor, manage, or learn more about their health through video, text, or interactive learning media (22-24). mHealth refers to portable or wireless applications such as text messaging, mobile-compatible applications (apps), wearable devices, or the use of social media (22, 23, 25). The integration of virtual care modalities into healthcare is quite promising due to their relatively low cost, scalability, and their association with successful patient engagement, satisfaction with communication with healthcare teams, and the facilitation of access to specialized care and diabetes monitoring (26, 27).

While focus on virtual interventions to improve glycemic control is important, this improvement may impact HRQOL as the responsibilities associated with this care may be burdensome for patients and their families (8). The eHealth and mHealth approaches increase patient involvement in their care and the need for familiarity with technical aspects of these care modalities (27, 28). There is currently no existing evidence synthesis of the impact of virtual interventions on HRQOL in children living with diabetes.

The objectives of this systematic review protocol to report on the methods of a systematic review to determine the effectiveness of eHealth and mHealth interventions in maintaining or improving HRQOL, glycemic control, overweight and obesity, and assess their impact on adverse events in children living with diabetes. Also, we aim to determine patient and caregiver satisfaction, and the acceptability and feasibility of virtual care modalities.

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Research questions

Primary: In children with diabetes, does the use of eHealth and mHealth interventions, when compared to in-person care, maintain or improve HRQOL?

Secondary: In children with diabetes, do virtual interventions maintain or improve glycemic control, body mass, and adverse diabetes-related outcomes when compared to in-person care? Is virtual care usable and acceptable for this population?

METHODS

The methodology for this protocol has been reported based on the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) guideline (29, 30) (Supplementary File 1). This protocol has been registered with PROSPERO (Registration Number: CRD42021235646). We aim to complete this systematic review between October-December 2021.

Patient and public involvement

Patients and/or the public were not involved in the design, conduct, reporting, or dissemination plans of this research.

Search Strategy

This review's search strategies will be developed by a Senior Health Sciences Librarian (LB), who has expertise in designing systematic review search strategies. A sample Ovid MEDLINE search strategy is reported in Table 1. Our searches will run on the following platforms: Ovid MEDLINE, Ovid Embase (Elsevier), Ovid EMCare (Elsevier), PsycInfo, and Web of Science (Clarivate). The grey literature searches will be done using ProQuest Dissertations and Theses A&I, and ClinicalTrials.gov.

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Table 1: Ovid MEDLINE Search Strategy

#	Searches
1	Child/
2	child*.mp.
3	Adolescent/
4	adolescen*.mp.
5	Pediatrics/
6	pediatric*.mp.
7	paediatric*.mp.
8	teen*.mp.
9	preadolescen*.mp.
10	preteen*.mp.
11	or/1-10
12	Diabetes Mellitus, Type 2/ or Diabetes Mellitus, Type 1/ or Diabetes Mellitus/
13	(T1D or T1DM or T2DM or T2D or NIDDM or IDDM).mp.
14	diabet*.mp.
15	or/12-14
16	Telemedicine/
17	telemedicine.mp.
18	tele-medicine.mp.
19	mhealth.mp.
20	ehealth.mp.
21	virtual health.mp.
22	Mobile Applications/
23	teleconsult*.mp.
24	web-based.mp.
25	mobile-based.mp.
26	Social media/
27	social media.mp.
28	twitter.mp.
29	telemonitor*.mp.

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30	Remote Consultation/
31	iphone.mp.
32	computers/ or computers, handheld/ or smartphone/
33	smart phone*.mp.
34	smartphone*.mp.
35	computer*.mp.
36	laptop*.mp.
37	((remote or web or virtual or online) adj2 (consult* or appointment*)).mp.
38	((web or virtual or online) adj2 (education or communit* or platform*)).mp.
39	ipad.mp.
40	(android adj3 (tablet* or device* or phone*)).mp.
41	or/16-40
42	Comparative Study/
43	comparative stud*.mp.
44	cohort studies/ or prospective studies/ or retrospective studies/
45	cohort stud*.mp.
46	Observational Study/
47	observation* stud*.mp.
48	nonrandom*.mp.
49	non-random*.mp.
50	Randomized Controlled Trial/ or Randomized Controlled Trials as Topic/
51	randomized controlled trial*.mp.
52	Random Allocation/
53	random*.mp.
54	Clinical Trial/
55	clinical trial.mp.
56	retrospective.mp.
57	prospective.mp.
58	or/42-57
59	11 and 15 and 41 and 58
60	remove duplicates from 59

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Inclusion Criteria

We will include studies that recruited boys and girls ages 2-18 years, from all ethnicities and diagnosed with T1DM or T2DM. The diagnosis of diabetes is stablished using standardized criteria, with a random or a 2-hour post oral glucose tolerance test ≥11.1 mmol/L (200 mg/dL), or a fasting glucose level of ≥7.0 mmol/L (126 mg/dL) (31).

Children at all pubertal stages based on Tanner staging (32, 33), glycemic control levels (using HbA1c), and Body Mass Index (BMI) z-scores will be included. Children treated with insulin injections of all insulin types and regimens (multiple daily injection regimens, twice-daily regimens, insulin mixes) will be included. We will also have children on insulin pump therapy regardless of pump type and those using glucometers or continuous glucose sensors as applicable.

We will include studies of all geographic locations, languages, and publication timing.

The eligible study designs to be included in this systematic review are randomized and non-randomized comparative studies. We will exclude protocol papers, case reports, and commentaries. Eligible studies which are not published in English will be translated using Google Translate.

Children with co-morbid autoimmune conditions (for example, Hashimoto's thyroiditis, Graves' disease, celiac disease), as well as children with diabetes-related complications including neuropathy, nephropathy, and retinopathy will be included.

This review will include interventions that deliver diabetes care via eHealth and mHealth approaches. Diabetes education and care refers to personalized patient and caregiver engagement to monitor and promote glycemic control and limit comorbidities (7, 10). eHealth refers to the Internet, virtual reality, or digital gaming applications that allow users to monitor, manage or

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learn more about their health through video, text, or interactive learning media (22-24). mHealth refers to portable or wireless applications such as text messaging, mobile-compatible applications (apps), wearable devices, or the use of social media (22, 23, 25). These applications also allow for user engagement with health monitoring, management, and learning (24, 25, 34). Comparator groups will be reported if available. These groups will be delivering treatment as usual; this approach includes diabetes education and care delivery via in-person clinic visits with diabetes healthcare teams. These visits include a review of glucose trends in logbooks or sensor downloads, a review of insulin regimens for injectors and pump settings for children on insulin pumps, dietary guidance, physical activity recommendations, and emotional and mental health screening and supports (15, 19). Also, information and education about lifestyle factors are included in these visits (15, 19).

We will include studies that report on the HRQOL, measured by diabetes HRQOL questionnaires. These can consist of the Pediatric Quality of Life Inventory (PedsQL) (35, 36) or the Diabetes Quality of Life for Youth Questionnaire (DQOL-Y) (37). Other questionnaires, such as the proxy formats of general HRQOL questionnaires for parents and families and others, will be included. Studies that have failed to report one or more measured outcomes will not be excluded.

Exclusion Criteria

We will exclude studies on boys and girls with other types of diabetes mellitus, including Maturity Onset Diabetes of the Young (MODY) and gestational diabetes mellitus, and cystic fibrosis-related diabetes, among others. Pregnant participants will also be excluded. Children receiving medications that can drive hyperglycemia e.g., steroids and immunosuppressants, will also be excluded.

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OUTCOMES MEASURES

Critical Outcome

This systematic review's critical outcome includes the maintained or improved HRQOL, as measured by diabetes HRQOL questionnaires, or proxy formats of the questionnaires used to measure HRQOL.

Important Outcomes

Important outcomes of this study include:

1) Maintained or improved glycemic control, as measured by HbA1c, logbook glucose readings, or glucose sensor-based data, e.g., DexCom G6™ (38), FreeStyle Libre sensors (39) or others. While the target HbA1c level for people with diabetes is usually <7.5% for T1DM and ≤ 6.5-7% for T2DM, goals that are adjusted to participants' needs will be accepted (6, 7, 40, 41).

2) Changes in body mass as measured by assessing BMI z-scores or BMI percentiles pre- and post-intervention changes.

3) The number and severity of morbidities. These morbidities include hypoglycemia (glucose levels ≤ 3.9 mmol/L (70 mg/dL)) and hyperglycemia (glucose levels ≥ 13.3 mmol/L (240 mg/dL)). Also, the occurrence of ketosis, assessed via measurement of blood beta-hydroxybutyrate or urine acetoacetate and using manufacturer-specific reporting measures including quantitative or semiquantitative testing as applicable. We will also assess rates of diabetic ketoacidosis (defined as hyperglycemia with blood glucose >11 mmol/L (200 mg/dL), venous blood gas-based pH <7.3 or serum bicarbonate <15 mmol/L, and the presence of ketones (≥3 mmol/L in blood, moderate-large ketonuria) (6, 42). If no specific values were reported to establish the diagnosis of hypoglycemia, hyperglycemia, ketosis, and diabetic ketoacidosis, the above events would be included as a dichotomous outcome.

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4) Patient and family satisfaction, and their perceptions of the usability and acceptability of the virtual intervention from relevant questionnaires, which include measures to assess frequency and ease of use and desire for continued application after the study has ended.

Data Management

We will export search results onto Covidence (43), remove duplicates, and complete data screening. Data will be abstracted into a Microsoft Excel sheet. We will develop and pilot test the data abstraction form to ensure its validity.

Study Selection

Two reviewers will independently assess the eligibility of titles, abstracts, and full-text articles identified in the database searches for potential eligibility and inclusion in this review. Disagreements at each stage will be resolved through discussion among the reviewers. A third reviewer will resolve persistent disagreements. The 2020 PRISMA flow diagram reported in Figure 1 will be utilized to track the process and will be included in the final paper (44, 45).

Data Abstraction

Data from included full-text articles will be abstracted independently by the two reviewers into the standardized data abstraction form. The reviewers will resolve any conflicts through discussion, and if necessary, a third reviewer will resolve any persisting discrepancies. We will abstract including first author's name, country, year, number of participants, age, sex, diabetes type, diabetes duration, comorbidities, insulin type and mode of delivery, and modality of glucose monitoring. Data on hyperglycemia, hypoglycemia, ketosis, and ketoacidosis will also be abstracted. We will abstract relevant data on the interventions including type, duration, details of delivery, HRQOL questionnaires used, HRQOL scores, glycemic control measures, change in glycemic control, change in BMI z-score or BMI percentile, adverse events, data on user

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satisfaction and the usability and acceptability of the interventions. Based on the data available, we will determine whether studies can be pooled for meta-analyses.

If there are missing data from eligible papers, we will e-mail the Principal Investigators.

Risk of Bias Assessment

Two reviewers will independently perform the risk of bias assessments of all randomized controlled trials using the Cochrane RoB 2.0 tool (46). Comparative, non-randomized studies will be assessed for risk of bias using the ROBINS-I tool (47). Any conflicts will be discussed, and a third reviewer will resolve any remaining disagreements.

GRADE assessment

To assess the overall confidence in the quality of evidence, we will use the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach (48). Quality will be determined by study design, risk of bias, inconsistency, indirectness, imprecision, and publication bias (48). The quality of the cumulative evidence and magnitude of the effect will be utilized to determine the overall strength of the meta-analysis findings if it is feasible to do so.

Statistical Analysis

We will include a summary table of all eligible studies in the final review. A meta-analysis will be conducted if two or more studies of similar study design, methods, populations, interventions, and outcomes are identified. We will not pool the data of randomized trials with those of non-randomized comparative studies. We will report dichotomous outcomes as risk ratios or odds ratios and continuous variables as mean differences or standardized mean differences, with a 95% confidence interval. A detailed qualitative synthesis of the findings will be provided in a narrative table if a meta-analysis is not possible.

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If a meta-analysis is possible, we will perform a random-effects model which accounts for within- and between-study variability. We will use the Review Manager (Revman 5.4) software to generate forest plots to represent the data graphically if appropriate (49).

Heterogeneity will be assessed using the χ^2 test p-value with a cut-off value of 0.1 and the I^2 statistic with a cut-off value of >75% defined as considerable heterogeneity, in addition to visual assessment of forest plots (50). A sensitivity analysis will be performed by removing studies with a high risk of bias to assess the impact on meta-analysis results (51).

If appropriate, we will conduct subgroup analyses on the type of diabetes mellitus, as participants of T1DM and T2DM have fundamentally different management goals. We will also perform subgroup analyses on the age and sex of the participants and the mode of insulin administration (injections versus pumps).

If ten or more studies are included, we will generate a funnel plot and utilize Egger's test to assess small study effect/publication bias. Alternatively, we will report publication bias by considering the number of relevant conference abstracts that did not have published articles.

DISCUSSION

While the number of children and youth living with diabetes is rising globally, the outcomes of these children have improved significantly over the past few decades, thanks to advances in care models and research (1, 7-12). However, the burden of care on patients and families is high due to the complex nature of the disease and its treatment, and burnout is a challenge that patients and families struggle with (13, 15, 16). With the covid-19 pandemic, new virtual models of care had to be adopted within a very short period of time to deliver care to these children and youth (17, 18). While the use of virtual care has grown steadily over the past few years, its impact on

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health-related quality of life with diabetes is less well understood compared to their impact on glycemic control. This systematic review will focus on understanding the impact of virtual diabetes care on HRQOL. As these care approaches now take an unprecedented priority in view of the pandemic, with healthcare systems and policy makers trying and balance safety with the need of delivering quality care worldwide, the data on HRQOL are crucial to help shape the future of virtual diabetes care.

As remote diabetes care and education have become viable alternatives to in-person care in the COVID-19 era, this systematic review will help develop an understanding of its effects on HRQOL outcomes in children with diabetes, and will guide future directives on implementing virtual diabetes care.

ETHICS AND DISSEMINATION

Ethics approval is not required as we will not recruit nor include any individual patient data in this systematic review. There are no other ethical and safety considerations. The results of our systematic review will be published in a peer-reviewed journal and presented at international conferences.

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Author's Contributions: MCS is the guarantor. RR, MK, LB, UA, LT, and MCS were responsible for developing the research question, study conception, and design. MCS, LB, RR, MK, UA, and LT developed the search strategy and eligibility criteria. RR and MCS were responsible for drafting the manuscript. All authors reviewed, edited, and approved the final version of the manuscript.

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Competing Interests: None declared.

Patient consent for publication: Not required.

Provenance and peer review: Not commissioned; externally peer reviewed.

Data availability statement: There are no data in this work.

Word Count: 2,539 words

Figure 1 Caption: Flow Diagram of Article Screening Process

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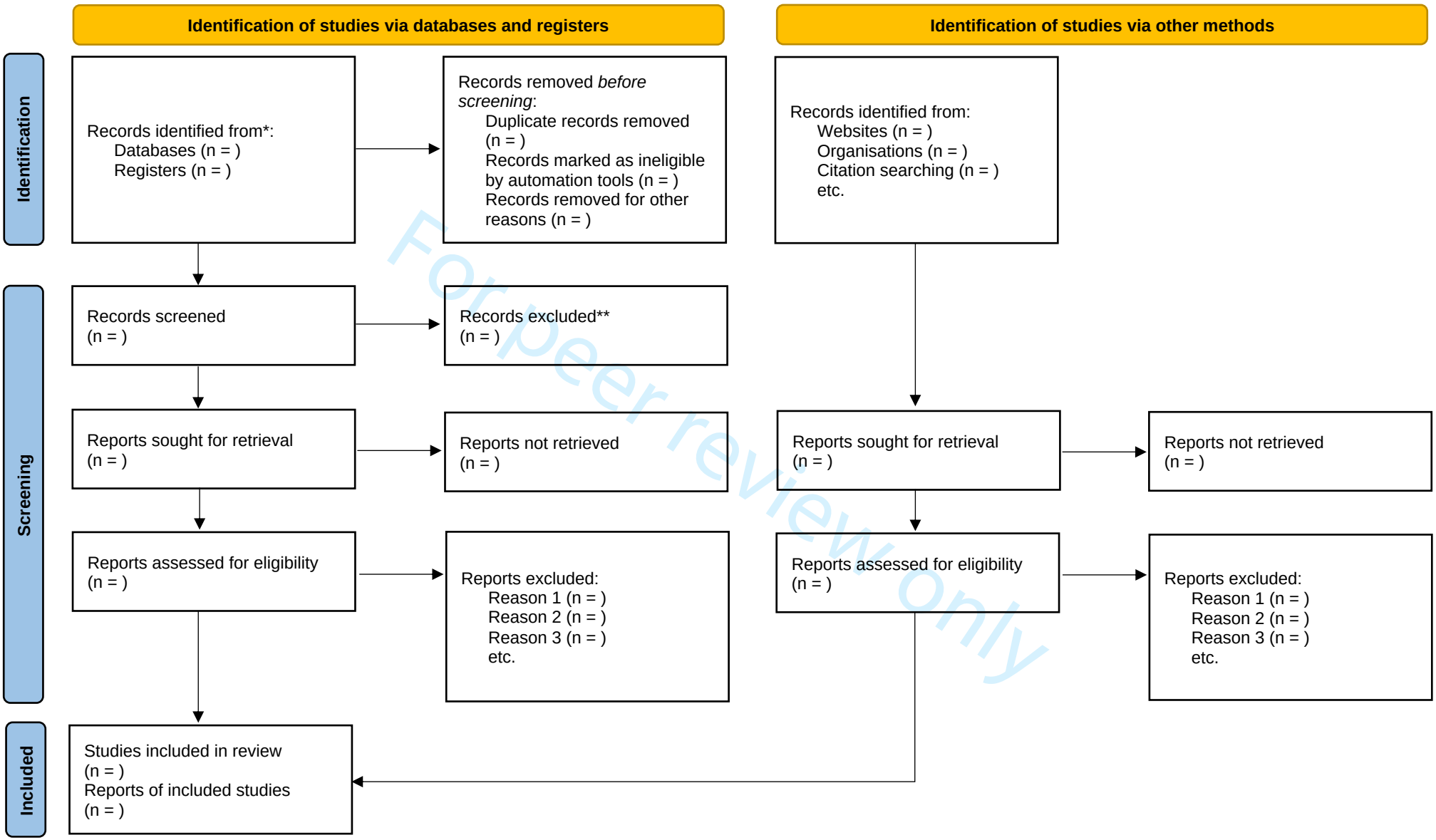
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Supplementary File 1: PRISMA-P Checklist

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1-2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	56, 140-141
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	4-37
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	314-318
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	N/A
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	319-320
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	319-320
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	319-320
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	117-122
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	123-127

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	154-199
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	147-152
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	153 (Table 1)
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	226-229
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	230-235
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	228-229, 236-248
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	239-246
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	200-225
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	249-253
DATA					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	261-266
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	268-273
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	273-278
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	266-267
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☒	☐	279-281
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	254-259

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol.

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